

Samar Kamboj

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EDUCATION

Indian Institute of Technology

Chennai, Tamil Nadu

Bachelors in Data Science – CGPA: 9.1

Expected - 2027

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Application Development, Linux

PROJECTS

E-Waste Logistics Platform

July 2026 – Aug 2026

- Built a lightweight e-waste logistics platform in a five member team; personally responsible for backend architecture, core API integrations, and CI/CD pipelines.
- Architected a **FastAPI**, **SQLAlchemy** backend and **JWT authentication** for secure role-based access; configured **CORSMiddleware** to securely manage cross-origin API traffic.
- Engineered an asynchronous task queue using **Redis** and **Celery** to offload email OTP generation and notification processing, paired with **webhooks** for low-latency, real-time in-app updates.
- Developed responsive user interfaces utilizing **React 19 (Vite 8)** and **Tailwind CSS**.
- Integrated the **Cashfree Payouts API** to automate UPI transfers.
- Built an AI support chatbot utilizing **LangChain** for real-time queries.
- Containerized the platform via **Docker** and configured one-click **Render** deployments with **PostgreSQL**.
- Managed version control with **Git** and Agile sprint planning via **Notion** within a modular codebase.
- Authored a robust **pytest** suite utilizing isolated database fixtures.

Music Genre Classification

Feb 2026 – Mar 2026

- Identified CPU-bound preprocessing as the primary training bottleneck; engineered a multi-GPU **PyTorch** pipeline with in-memory data caching and on-GPU audio-to-spectrogram transformation with SpecAugment, reducing end-to-end training time by ~95%.
- Developed and fine-tuned an Audio Spectrogram Transformer (AST) end-to-end, achieving a **Macro-F1** score of **0.923** and deployed the model on **Hugging Face** for scalable, real-time audio genre inference.

Cinema Audience Forecasting

Oct 2025 – Nov 2025

- Built a demand forecasting system using **OLS**, **XGBoost**, **LightGBM** within **Scikit-learn** pipelines, engineering cyclical datetime features (sine/cosine) and autoregressive lag variables with **Pandas**, **NumPy**, and **Matplotlib** to capture seasonal and weekly attendance patterns.
- Designed an iterative multi-horizon prediction algorithm that feeds prior-day forecasts into future time steps, with hyperparameters tuned via **TimeSeriesSplit** cross-validation.

TECHNICAL SKILLS

Languages: Python, SQL

Data Science: Pandas, NumPy, Scikit-learn, Statsmodels, PyTorch, RAG

Backend & Databases: FastAPI, SQLAlchemy, MySQL, PostgreSQL, Redis

Developer Tools: Git/Github, Docker, Celery, pytest

CERTIFICATES

- GATE CS: Qualified in top 5%
- NPTEL: Getting Started with Competitive Programming (93%)
- HackerRank Advanced SQL